# Active Learning for Adaptive Grasp Speed Selection

## End-to-end feasibility results and research limitations

**Engineering feasibility demonstrated; research advantage not yet demonstrated.** The system collected physical success/failure labels, trained a classifier, saved three-speed predictions, and executed its recommendations in MuJoCo. This is not an active-learning comparison or evidence that adaptive speed selection beats a fixed fast setting.

## Research question

Can active learning use fewer grasp trials than random sampling to learn an equally effective or better speed-selection policy? Level 1 checks whether the physical data and execution pipeline works before that hypothesis is tested.

## Environment and task

MuJoCo 3.3.7 with a Franka Panda arm and gripper. The existing black-bowl/drawer workcell supplies two concrete targets: a black bowl and a wine bottle. Each trial performs approach, descent, closure, lift, and hold. **Drawer placement/closure and six-object sorting are not part of this classifier experiment.**

## Collected labels

| Split | Attempts | Success | Failure | Invalid setup |
|---|---|---|---|---|
| Training | 108 | 81 | 18 | 9 |
| Evaluation | 54 | 48 | 6 | 0 |

Invalid initial states have no label. Expected controller failures remain negative examples; internal/numerical errors abort instead of becoming fabricated failures.

## Observed collection cost

| Measurement | Observed result |
|---|---|
| Valid-label trials | 153 trials; 2.189 wall s/trial |
| Complete study function | 338.58 wall s |
| Classifier training | 0.043 wall s |

Measured on Apple M5 Pro / 48 GB, headless, with a separate live preview. Per-trial timing includes construction, settling, execution, hold observation, trace saving and cleanup; it excludes the final row-JSON write. These are wall-clock costs, not robot task durations. A larger collection campaign has not yet been measured.

# Features, model, and physical verification

| Input | Definition in this pilot |
|---|---|
| Object category | Two one-hot columns: bowl and bottle |
| Initial position | Prescribed reset X/Y offsets, each within +/-0.08 m |
| Initial orientation | Prescribed reset yaw within +/-0.20 rad |
| Descent speed | 0.04, 0.10, 0.20 m/s trajectory settings |

The six columns are category (2), reset X/Y (2), yaw (1), and speed (1). Actual settled poses are archived separately, not used as these input features. There is no visual recognition. Speed is a trajectory parameter, not a measured constant TCP speed.

Each object's geometry, mass, friction, and grasp rule stay fixed. Initial opening is 0.08 m for the bowl and 0.04 m for the bottle; closure-trigger height is 0.

## Split and execution order

Seed 5644: 36 training conditions and 18 held-out initial conditions, each tried at three speeds. Both splits use the same two object assets: this is not unseen-object generalization. The three outcomes within a condition are correlated.

Predictions and recommendations were saved before any evaluation execution. The recommended speed was executed first, then the alternatives from identical initial-state fingerprints. Evaluation labels were not used for retraining.

## Predictor and selection rule

Random forest: 128 trees, maximum depth 6, one worker. Select the fastest speed with raw predicted probability at least 0.9. Otherwise select the highest score, breaking ties toward the slower speed, and mark the threshold unmet. This pilot uses neither GPT, a 27B action model, nor MRIoC to choose speed.

| Held-out condition | 0.04 | 0.10 | 0.20 | Chosen m/s | Observed |
|---|---|---|---|---|---|
| bowl.evaluation_00 | 1.0 | 1.0 | 1.0 | 0.20 | Success |
| bottle.evaluation_02 | 0.0 | 0.0 | 0.0 | 0.04 * | Failure |
| bottle.evaluation_08 | 0.0 | 0.0 | 0.0 | 0.04 * | Success |

* Threshold unmet; execution still followed the predeclared simulation policy. These probabilities are uncalibrated, not safety guarantees. The final row is an explicit wrong prediction: a zero-score grasp actually succeeded.

## A positive label requires physical evidence

Lift at least 0.10 m, bilateral grip, table clearance and stable relative pose for 2 continuous simulation seconds. An independent observer checks every 2 ms; traces are saved every 20 ms and at phase changes. The older controller's one-second success flag alone does not suffice.

# Task performance and interpretation

| Policy | Successes | Mean successful-task duration (sim s) |
|---|---|---|
| Fixed 0.04 m/s | 16/18 | 18.642 |
| Fixed 0.10 m/s | 16/18 | 14.138 |
| Fixed 0.20 m/s | 16/18 | 12.634 |
| Random-forest recommendation | 16/18 | 13.010 |

Durations are success-only simulation time, not video length or collection wall time. Here, the same 16 conditions succeeded under all four policies.

The learned recommendation was **0.376 simulation seconds slower** than the fixed fastest setting on joint successes. Across both splits, every valid condition had the same success/failure label at all three speeds. No fastest-fails/slower-succeeds case was observed. There is therefore no demonstrated adaptive-speed benefit.

Descriptive prediction accuracy: 94.44%; Brier score: 0.05556. Classification accuracy is not evidence of faster or more reliable speed selection. Probabilities remain uncalibrated.

## Failures and safety boundary

Training failures: 15 descent-arrival failures, one lift timeout and two hold failures. All six evaluation failures were descent-arrival failures. Unexpected contacts occurred in nine evaluation trials, including three successful grasps. No numerical warnings were recorded. Success does not imply collision-free operation or real-robot readiness.

## Answers to the professor's three questions

**1. Is the simulator ready and fast enough?** Yes for a small classifier feasibility study: the measured batch completed real-label collection, training and recommended physical execution. The environment is already operational rather than a proposed build.

**2. What changes besides speed?** Object category, reset X/Y and yaw. Per-object physical parameters and grasp rules stay fixed. The scope is two existing assets and held-out initial states, not arbitrary new objects.

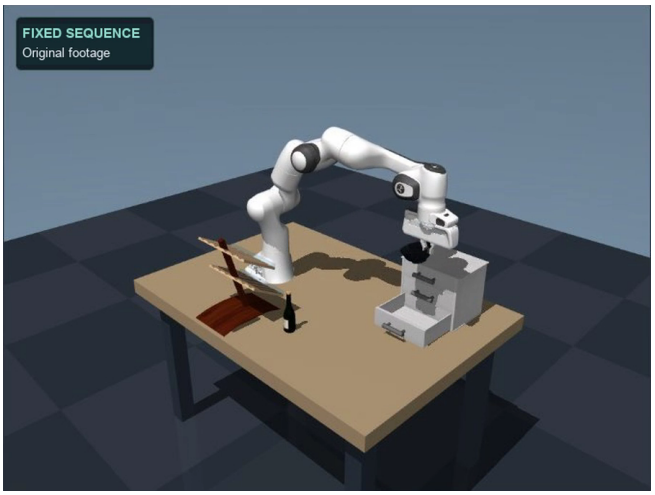
**3. Why not D-optimal?** It is a legitimate candidate control, not something this pilot rules out. Active learning, random sampling and D-optimal design have not yet been compared. A fair comparison requires a specified model/design criterion and the same initial data, label budget and independent evaluation set.

## Next research decision

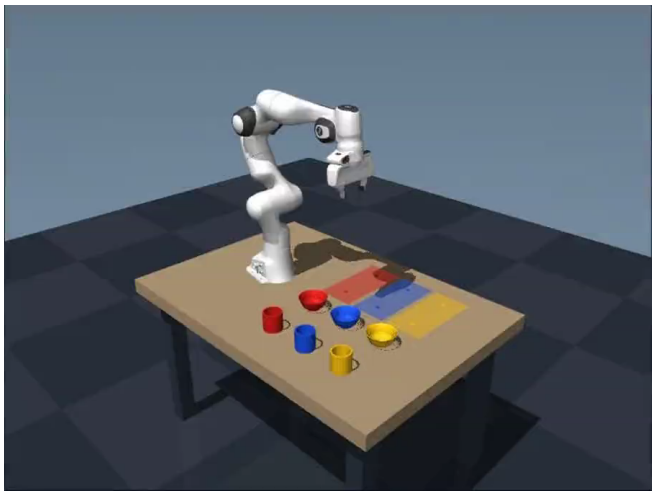
First verify how the speed settings affect actual trajectories and seek reasonable, predeclared conditions with meaningful success/time tradeoffs. Do not manufacture negative labels. Only then test learning curves and task performance. If speed choice remains trivial, revise the question with the professor rather than claiming an advantage.

# Demonstrations and reproducible evidence

These two independent historical recordings show the broader prototype. Their English editions retain the original motion and presentation timestamps; they are not new simulation runs or new GPT requests. **Neither is a classifier evaluation trial.**




**Black-bowl storage.** Pick, place in the drawer, close it, return home. Frame at 36.00 s; one approved proposal followed by a fixed local sequence.



**Six-object sorting.** Three bowls and three cups. Frame at 20.00 s; one approved GPT queue followed by local contact-feedback execution.

English storage video | English sorting video

## Reproduce the Level 1 experiment

```
uv sync --locked
uv run --locked python run_grasp_learning.py \
  --output results/grasp_level1_reproduction
```

Run from the repository root; the output directory must not exist. Python 3.12.13, MuJoCo 3.3.7, scikit-learn 1.9.1. Only load joblib files you trust.

## Evidence and provenance

`results/grasp_level1_verified_20261104/`

Frozen protocol | Environment / fingerprints | Summary | Predictions | Executed outcomes | Detailed record

Raw trials, compressed physical traces, training/evaluation data and the saved model are retained alongside these files. This verified batch replays the earlier batch's frozen protocol; it is not an independent dataset. Do not pool the two batches into 324 independent samples.

Summary SHA-256:
`7374a1d07a94e2b7d96f6c60cfc5d90e9310bcacc771b22cb4146e4befd3b463`

## Verification scope

All 34 added Level 1 checks passed in the initial 319-test run. That run also had nine errors across three historical sorting methods/subtests due to fingerprint mismatch. After test-fixture fixes, nine relevant protocol/physics tests and one server-layout test passed; the complete suite was not rerun. Original logs remain archived.

This report regenerates from archived evidence via `uv run --script build_grasp_speed_report.py`. Linked repository artifacts require GitHub access while the repository is private.